



MALLA REDDY UNIVERSITY

M.Tech I Year – II Semester

Minor – II Examination

Subject Name: Deep Learning & Its Applications

Subject Code: MR25-2PG0107

UNIT – III

Q.No	Question	Marks
Q1.	Explore the concept of dataset augmentation in deep learning and how it contributes to improving model generalization	
Q2.	What is the role of denoising autoencoders in deep learning, and how do they enhance the robustness of a model?	

UNIT – IV

Q3.	Explain the concept of greedy layer wise pre-training, and how it is used to improve the training of deep neural networks?	
Q4.	How does Batch Normalization work, and how can we use this technique to improve the training stability and generalization performance of Deep Learning models?	
Q5.	How does the Word2Vec algorithm contribute to learning vector representations of words? Explain the underlying principles of continuous bag-of-words (CBOW) and skip-gram models within Word2Vec.	

UNIT – V

Q6.	What are Gated Recurrent Units (GRUs) and Long Short-Term Memory (LSTM) cells, and how can these specialized RNN architectures be used to model long-term dependencies and avoid the vanishing gradient problem?	
Q7.	Explore the GoogLeNet architecture and elucidate some of its significant contributions to the field of deep learning	
Q8.	Explore the applications of Deep Dream and Deep Art in the field of deep learning. How do these techniques leverage neural networks to generate visually captivating and artistic images?	



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M.Tech I Year – II Semester
Minor – II Examination
Subject Name: Generative AI & Its Applications
Subject Code: MR25-2CB0108

UNIT – III

Q.No	Question	Marks
Q1.	Critically compare encoder-only, decoder-only, and encoder-decoder Transformer architectures. Justify the architectural choice for three distinct NLP tasks.	8
Q2.	Evaluate techniques for extending the Transformer context window (sliding-window attention, sparse attention, RoPE interpolation), analyzing the trade-offs in compute and accuracy.	8

UNIT – IV

Q3.	Explain the Retrieval-Augmented Generation (RAG) architecture. Analyze how retrieval quality affects generation quality and propose strategies to improve retrieval precision.	8
Q4.	Evaluate hallucination-mitigation techniques (grounding, RAG, fact verification, self-consistency). Justify which combination is most robust for a high-stakes application.	8
Q5.	Analyze prompt injection attacks on LLM-based applications. Evaluate at least three mitigation strategies and discuss their trade-offs.	8

UNIT – V

Q6.	Discuss and evaluate the Stable Diffusion architecture (U-Net, VAE, CLIP text encoder), analyzing how cross-attention conditions image generation on text prompts.	8
Q7.	Explain deepfake detection techniques, analyzing the ongoing arms race between generation and detection methods.	8
Q8.	Discuss emerging trends in Generative AI - agentic AI, multimodal foundation models, and AI regulation. Predict the trajectory of the field over the next five years, justifying your reasoning.	8

Subject Name: INTERNET OF THINGS
Subject Code: MR25-2CB0048

UNIT - III

Q.No	Question	Marks
Q1.	Explain & Analyze the SPI communication protocol and implement an example Arduino program to interface an SPI-based module (e.g., SD card or display) with justification of the wiring.	
Q2.	Evaluate the role of built-in and third-party Arduino libraries in simplifying sensor interfacing and implement an example using a library-based sensor (e.g., DHT11) program.	

UNIT - IV

Q3.	Analyze the hardware architecture and GPIO layout of Raspberry Pi and compare it with Arduino Uno in terms of processing capability for IoT applications.	
Q4.	Design and implement a Python program to interface a sensor (e.g., DHT11 or ultrasonic sensor) with Raspberry Pi and analyze the data acquisition process.	
Q5.	Implement and analyze the process of remotely accessing and programming Raspberry Pi using SSH, including security considerations for IoT deployment	

UNIT - V

Q6.	Design and implement an integration of Raspberry Pi with a cloud platform (e.g., ThingSpeak/AWS IoT) for real-time IoT data logging, and analyze the end-to-end data flow.	
Q7.	Critically evaluate the advantages and limitations of implementing IoT in agriculture, supporting your analysis with a practical implementation example.	
Q8.	Design and analyze the architecture of a Healthcare IoT system with a neat diagram, implementing an example of a remote patient-monitoring scenario.	

Subject Name: Large Language Models

Subject Code: MR25-2CB0045

UNIT - III

Q.No	Question	
Q1.	Explain supervised fine-tuning (SFT) process and datasets used.	
Q2.	Compare RLHF with Constitutional AI.	

UNIT - IV

Q3.	Explain zero-shot, few-shot, and chain-of-thought prompting with examples.	
Q4.	Explain vector databases and their role in LLM applications.	
Q5.	Explain how to build a production-grade RAG system using LangChain.	

UNIT - V

Q6.	Explain LLM agents and their components such as tools, memory, and planning.	
Q7.	Discuss real-world applications of LLMs in healthcare and finance.	
Q8.	Explain ethical issues, data privacy, and governance in LLM deployment.	

Subject Name: COMPUTER VISION AND ITS APPLICATIONS

Subject Code: MR25-2CB0046

UNIT - III

Q.No	Question	
Q1.	Apply Median filtering to remove salt-and-pepper noise from an image.	
Q2.	Evaluate the effectiveness of sharpening techniques in enhancing image quality.	

UNIT - IV

Q3.	List different threshold selection methods.	
Q4.	Apply Otsu's thresholding method to segment an image.	
Q5.	Analyze the effect of illumination changes on threshold selection.	

Q6.	State the principle of the Canny edge detector.	
Q7.	Analyze the influence of threshold values on edge detection performance.	
Q8.	Describe hysteresis thresholding in the Canny edge detector	

Subject Name: Data Analytics using Python
Subject Code: MR25-2CB0047

UNIT - III

Q.No	Question	
Q1.	Discuss the concept of Excel Styles, illustrating your answer with an example, and explain how they contribute to effective and consistent spreadsheet formatting.	
Q2.	Demonstrate with examples how to create, apply, and modify cell styles to enhance the readability and consistency of large data sets.	

UNIT - IV

Q3.	Explain how pivot charts complement pivot tables with visual representation of data	
Q4.	Compare and contrast different chart types (Column, Line, Pie, and Bar). Discuss scenarios where each chart type is most effective.	
Q5.	Evaluate how using Slicers improves interactivity and data visualization in PivotTables and PivotCharts. Provide examples	

UNIT - V

Q6.	Demonstrate how to restrict and sort data in SQL using WHERE, ORDER BY, and DISTINCT clauses with appropriate examples.	
Q7.	Discuss SQL joins for displaying data from multiple tables with an example	
Q8.	Explain Data Manipulation Language (DML) statements — INSERT, UPDATE, DELETE, and MERGE. Provide syntax and examples for each.	

Subject Name: Quantum Computing

Subject Code: MR23-2BS0131

Important Questions

UNIT – III

Q.No	Question	
Q1.	Explain the concept of superposition in quantum computing and its significance for Qubits.	
Q2.	Describe the Pauli-X, Y and Z gates and its role in quantum computing.	

UNIT – IV

Q3.	Explain the concept of quantum parallelism and how it is utilized in Deutsch's algorithm.
Q4.	What is the main challenge in classical factoring algorithms, and how does Shor's algorithm overcome it?
Q5.	What is quantum annealing and how it relates to finding the ground state of a system.

UNIT – V

Q6.	Write the Qiskit code to create a simple quantum circuit that places a single qubit in an equal superposition state and then measures it.
Q7.	Explain implementation of Toffoli Gate using Qiskit
Q8.	Outline the steps to set up an IBM Quantum Experience account